Smart Home Demo Lab Team 2 Update 4 Report

Yixue Wang, Linghao Dong, Yang Xiao, Chenxu Liu, Tong Lu

December 10, 2025

Overview of Work in the Past Two Weeks

Application Layer

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  Device registration, enable/disable, state sync.

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- **Hardware Communication**
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- **Logging Module**
  Unified logging configuration.

Demo Video

Click to Play Demo Video

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Unified management of smart device lifecycle, providing registration, state sync, and control capabilities.

Design Concept: Acts as a mediator, encapsulating device management flows, coordinating domain objects, and publishing events.

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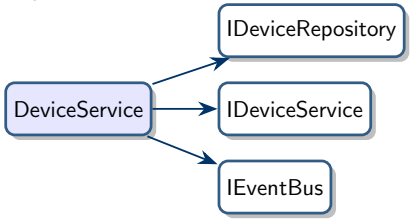

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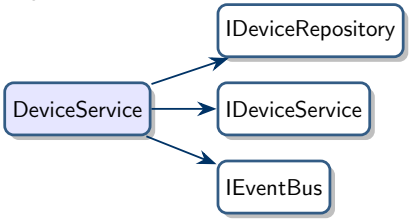


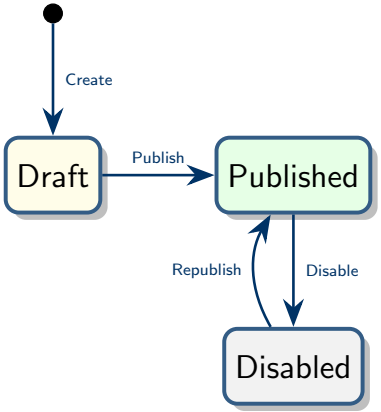
Functional List

Method	Description
`register_device()`	Register, generate UUID, publish event
`enable/disable`	Toggle state, publish status change
`sync_device_state()`	Delegate domain service to sync state
`get_device(_by_id)`	Get device details
`list_devices()`	Query list (with filters)
`delete_device()`	Delete device

Application Layer - Scene Service

Manages the full lifecycle of scenes and validates scene definitions.

Scene State Machine



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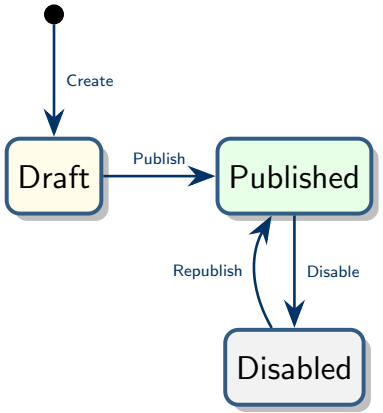


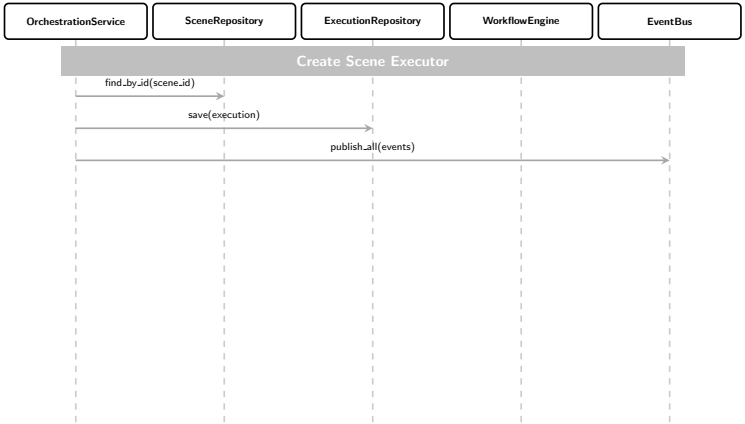
Functional List

Method	Description
`create_scene()`	Create draft scene
`update_scene(_def)`	Update info/def (w/ validation)
`publish_scene()`	Publish (triggers event)
`disable_scene()`	Disable (triggers event)
`get_scene(_def)`	Get details/definition
`list_scenes`	Query list
`delete_scene()`	Delete scene

Application Layer - Orchestration Service

Acts as the coordinator of three contexts, triggers execution, and drives the workflow.

Execution Flow



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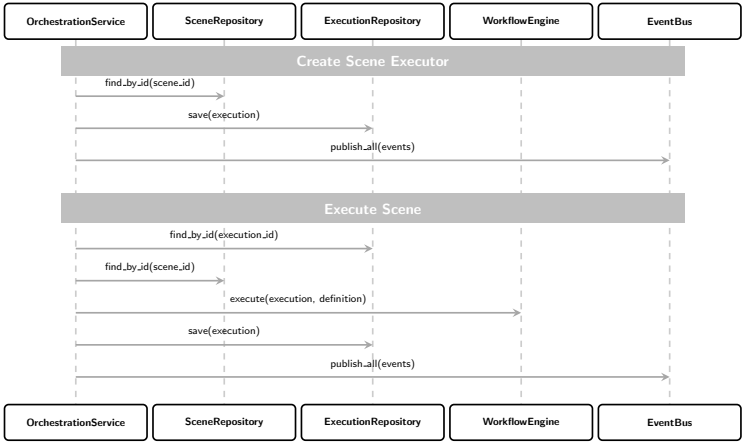

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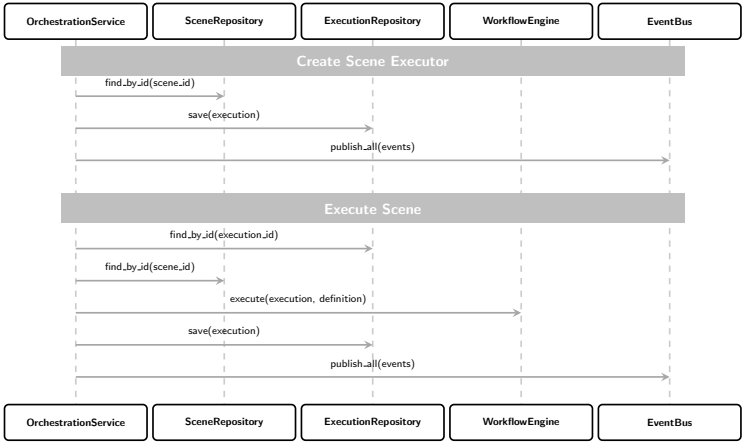

Functional List

Method	Description
`trigger_execution()`	Triggers execution, creates record
`execute_scene()`	Invokes engine to execute scene
`trigger_and_execute()`	Trigger & execute immediately
`get_execution()`	Retrieves execution record
`get_execution_details()`	Gets structured exec details
`list_executions()`	Queries execution list
`list_executions_for_scene()`	Gets all executions for a scene

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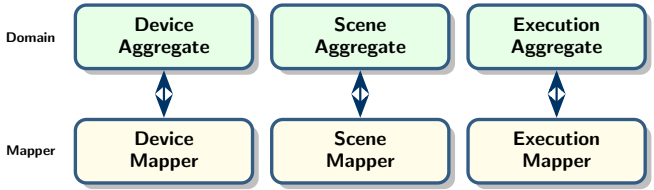

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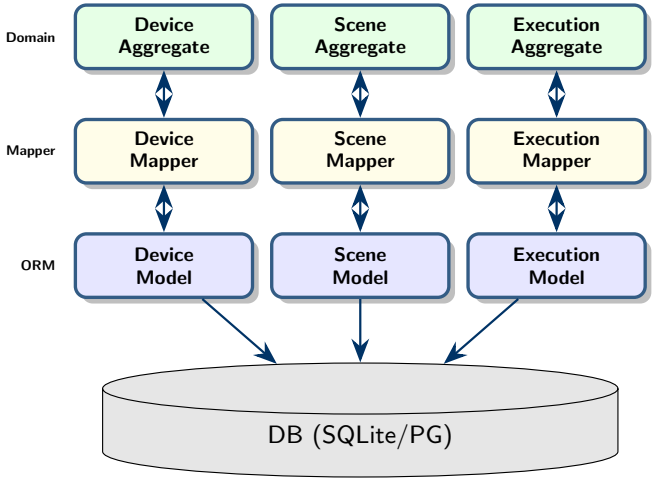

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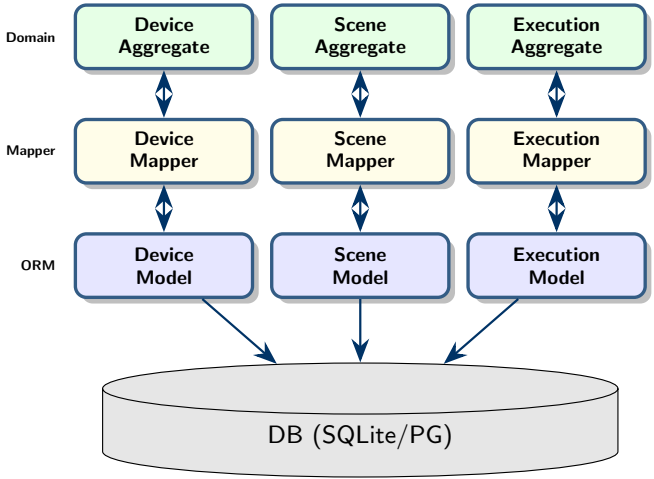

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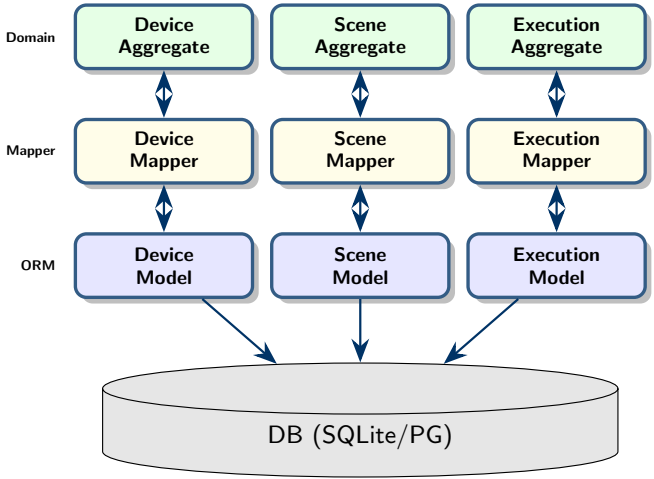

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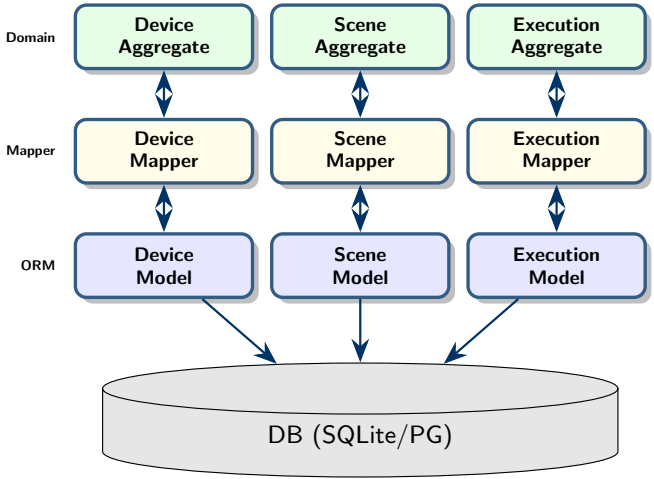

Key Features

- **Isolation**: Strict boundaries.
- **Mapping**: Domain ↔ ORM.
- **Repository**: Encapsulates data access.

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Decouples context communication, based on asyncio.PriorityQueue.

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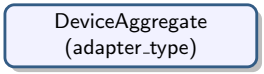
Core Interface Definition

```
class IEventBus:
    # Publish single event
    async def publish(event, priority)

    # Publish batch
    async def publish_all(events)

    # Subscribe to event type
    def subscribe(event_type, handler)

    # Unsubscribe
    def unsubscribe(event_type, handler)
```

Infrastructure Layer - Hardware Communication

Abstracts platform differences, provides unified control interface.

Layered Design Architecture



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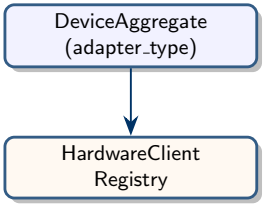

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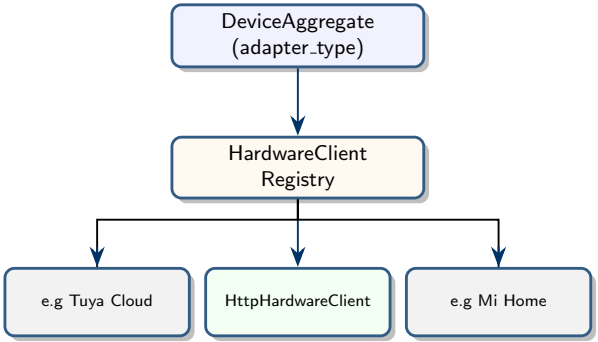

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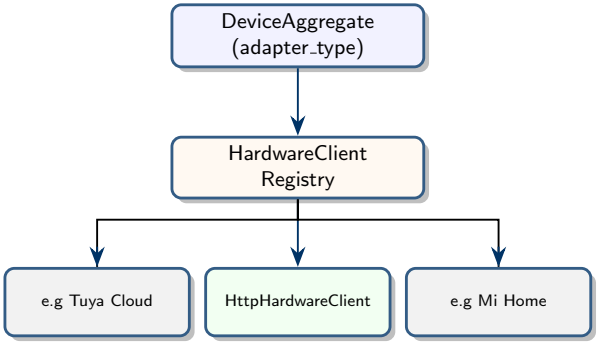

HttpHardwareClient

Feature	Description
Platform	API
Auth	Token
Methods	`call`, `get`

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